

KENDRIYA VIDYALAYA SANGATHAN MUMBAI REGION
PRE –BOARD 1 (SESSION : 2024-25)
MATHEMATICS – STANDARD (041)

Time Allowed: 3 Hours

Maximum Marks :80

General Instructions:

Read the following instructions carefully and follow them:

1. This question paper contains 38 questions.
2. This Question Paper is divided into 5 Sections A, B, C, D and E.
3. In Section A, Questions no. 1-18 are multiple choice questions (MCQs) and questions no. 19 and 20 are Assertion- Reason based questions of 1 mark each.
4. In Section B, Questions no. 21-25 are very short answer (VSA) type questions, carrying 02 marks each.
5. In Section C, Questions no. 26-31 are short answer (SA) type questions, carrying 03 marks each.
6. In Section D, Questions no. 32-35 are long answer (LA) type questions, carrying 05 marks each.
7. In Section E, Questions no. 36-38 are case study based questions carrying 4 marks each with sub parts of the values of 1, 1 and 2 marks each respectively.
8. All Questions are compulsory. However, an internal choice in 2 Question of Section B, 2 Questions of Section C and 2 Questions of Section D has been provided. An internal choice has been provided in all the 2 marks questions of Section E.
9. Draw neat and clean figures wherever required.
10. Take $\pi = 22/7$ wherever required if not stated.
11. Use of calculators is not allowed.

SECTION – A (MCQ)

(1x10 = 10)

Q.1. If the HCF of 85 and 153 is expressible in the form of $85n-153$, then the value of n is

- (a) 3 (b) 2 (c) 4 (d) 1

Q.2. The quadratic polynomial, the sum and product of whose zeroes are -1 and 1 respectively, is

- (a) $x^2 - 1$ (b) $x^2 + 1$ (c) $x^2 + x + 1$ (d) $x^2 - x$

Q.3. If the system of equations $3x+y=1$ and $(2k-1)x+(k-1)y=2k+1$ is inconsistent, then k =

- (a) -1 (b) 0 (c) 1 (d) 2

Q.4. The quadratic equation $2x^2 - \sqrt{5}x + 1 = 0$ has

- (a) 2 distinct real roots (b) 2 equal real roots
(c) No real roots (d) more than 2 real roots

Q.5. The discriminant of the equation $bx^2 + ax + c = 0$, $b \neq 0$ is given by

- (a) $\sqrt{b^2 - 4ac}$ (b) $\sqrt{a^2 + 4bc}$ (c) $\sqrt{a^2 - 4bc}$ (d) $\sqrt{b^2 + 4ac}$

Q.6. The value of x for which $2x$, $(x+10)$ and $(3x+2)$ are three consecutive terms of an AP is
 (a) 6 (b) 0 (c) 1 (d) 3

Q.7. If $\triangle ABC \sim \triangle DEF$ then, which of the following is true?

(a) $BC.EF = AC.FD$ (b) $AB.ED = AC.DE$ (c) $BC.DE = AB.EF$ (d) $BC.DE = AB.FD$

Q.8. In $\triangle ABC$, D and E are points on the sides AB and AC respectively such that $DE \parallel BC$.

If $\frac{AD}{DB} = \frac{2}{3}$ and $AC = 18$ cm, then AE is equal to

(a) 5.2 cm (b) 6.2 cm (c) 8.2 cm (d) 7.2 cm

Q.9. If 'a' is any positive integer such that the distance between the points $P(a, 2)$ and $Q(3, -6)$ is 10 units, then the value of 'a' is

(a) -3 (b) 9 (c) 6 (d) 3

Q.10. Cards numbered 7 to 40 were put in a box. Anish selects a card at random. What is the probability that the selected card is a multiple of 7?

(a) $\frac{7}{34}$ (b) $\frac{5}{34}$ (c) $\frac{6}{35}$ (d) $\frac{7}{35}$

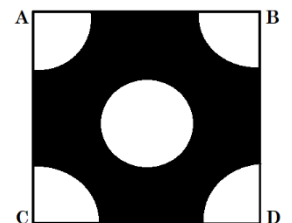
Q.11. $\frac{\sin \theta}{\sqrt{1 - \sin^2 \theta}}$ can also be written as

(a) $\cot \theta$ (b) $\sqrt{\sin \theta}$ (c) $\frac{\sin \theta}{\sqrt{\cos \theta}}$ (d) $\tan \theta$

Q.12. If the radius of the circle is 13 cm and the chord is 10 cm, then the length of the perpendicular drawn from the centre to the chord is

(a) 12 cm (b) 13 cm (c) 8 cm (d) None of these

Q.13. From each corner of a square of sides 4 cm a quadrant of a circle of a radius 1 cm is cut and also a circle of a diameter 2 cm is cut. The area of the remaining portion of the square is (see the figure, use $\pi = 3.14$)



(a) 10.25 cm^2 (b) 9.72 cm^2
 (c) 11.52 cm^2 (d) None of these

Q.14. Diameter of a sphere is 6 cm. It is melted and drawn into a wire of radius 0.2 cm. Then the length of the wire is

(a) 6 cm (b) 700 cm (c) 900 cm (d) None of these

Q.15. If we join two hemispheres of same radius along their bases, then we get a:

(a) Cone (b) Cylinder (c) Cuboid (d) Sphere

Q.16. Weight of 40 eggs were recorded as given below.

Weights (in gm)	85-90	90-95	95-100	100-105	105-110
No. of eggs	10	12	12	4	2

The upper limit of the median class is

(a) 90 (b) 95 (c) 94.5 (d) 89.5

Q.17. Using the empirical formula, the mode of a distribution whose mean is 8.32 and median is 8.05

- is (a) 8.51 (b) 7.51 (c) 16.37 (d) 40.79

Q. 18. If red face cards are removed from the deck of 52 playing cards, then the probability of getting a black jack is

- (a) $\frac{2}{46}$ (b) $\frac{2}{52}$ (c) $\frac{4}{48}$ (d) $\frac{2}{23}$

In the following questions 19 and 20, a statement of assertion (A) is followed by a statement of reason (R). Mark the correct choice as:

- (a) Both assertion (A) and reason (R) are true and reason (R) is the correct explanation of assertion (A).
 (b) Both assertion (A) and reason (R) are true but reason (R) is not the correct explanation of assertion (A).
 (c) Assertion (A) is true but reason (R) is false.
 (d) Assertion (A) is false but reason (R) is true.

Q.19. **Assertion (A)** : The point which divides the line segment joining the points A(1, 2) and B(-1, 1) internally in the ratio 1:2 is $\left(-\frac{1}{3}, \frac{5}{3}\right)$.

Reason (R) : The coordinates of the point which divides the line segment joining the points $A(x_1, y_1)$ and $B(x_2, y_2)$ in the ratio $m_1 : m_2$ are $\left(\frac{m_1 x_2 + m_2 x_1}{m_1 + m_2}, \frac{m_1 y_2 + m_2 y_1}{m_1 + m_2}\right)$.

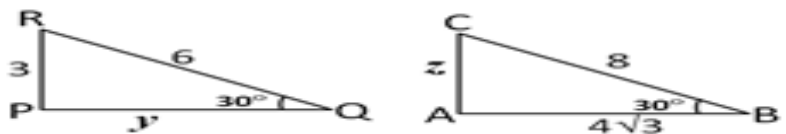
Q.20. **Assertion (A)** : If $\sin A = \frac{1}{3}$, ($0^\circ < A < 90^\circ$), then the value of $\cos A$ is $\frac{2\sqrt{2}}{3}$.

Reason (R) : For every angle θ , $\sin^2 \theta + \cos^2 \theta = 1$.

SECTION – B (VSA)

(2x5 = 10)

Q.21. In the figure $\triangle ABC \sim \triangle PQR$, find value of $y + z$.



Q.22. If $12 \operatorname{cosec} A = 13$, then find the value of $\frac{2\sin A - 3\cos A}{4\sin A - 9\cos A}$

Q.23. Area of a sector of a circle of radius 36 cm is $54\pi \text{ cm}^2$. Find the length of the corresponding arc of the sector.

Q.24. Amandeep took a pack of 52 cards. She kept aside all the black face cards and shuffled the remaining cards well. Write the probability of getting Queen card.

OR

A bag contains 100 cards numbered 1 to 100. A card is drawn at random from the bag. What is the probability that the number on the card is a (i) perfect square? (ii) perfect cube?

Q.25. The LCM of two numbers is 14 times their HCF. The sum of LCM and HCF is 750. If one number

is 250, then find the other number

OR

Ravi, Seema, and Raju start preparing cards for all persons in an old age home. They complete one card in 20, 16 and 10 minutes. If all of them started together at the same time, after what time will they start preparing a new card together?

SECTION – C (SA-II)

(3x6 = 18)

Q.26. Prove that $\sqrt{5}$ is irrational.

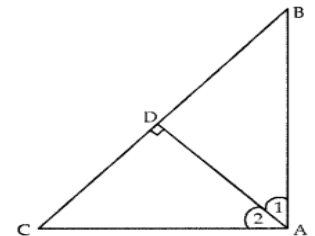
Q.27. Find the zeroes of the polynomial $4t^2 - 4t + 1$ and verify the relationship between the zeroes and the coefficients of the polynomial?

Q.28. The sum of two numbers is 9 and the sum of their reciprocals is $\frac{1}{2}$. Find the numbers?

Q.29. Prove the following identities: $(\sin \theta + \operatorname{cosec} \theta)^2 + (\cos \theta + \sec \theta)^2 = 7 + \tan^2 \theta + \cot^2 \theta$

Q.30. In a $\triangle ABC$, if $\angle A = 90^\circ$ and $AD \perp BC$,

Prove that $AD^2 = BD \times DC$.



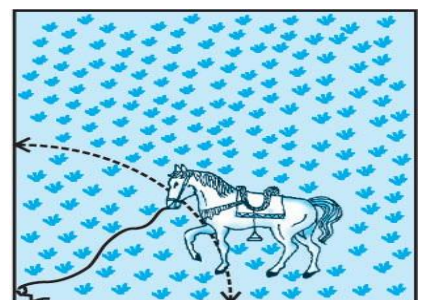
OR

A vertical pole of length 8 m casts a shadow 6 m long on the ground and at the same time a tower casts a shadow 30 m long. Find the height of tower.

Q.31. A horse is tied to a peg at one corner of a square shaped grass field of side 15 m by means of a 5 m long rope.

(See the attached figure)

- (i) The area of that part of the field in which the horse can graze.
- (ii) The increase in the grazing area if the rope were 10 m long instead of 5 m. (Use $\pi = 3.14$)



OR

Rahul bought a pair of glasses with wiper blades. He was curious to know the area being cleaned by each of wiper blades. With the help of ruler and a protractor, he found the length of each blade as 3 cm and the angle swept as 60° .

- (i) Find the area that each wiper cleans in one swipe, in terms of π .
- (ii) If the diameter of each circular glass is 5 cm, what percent of the area of the glass will be cleaned by the blade in one swipe?

SECTION – D (LA)

(5x4=20)

Q.32. The median of the following data is 525. Find the values of x and y, if the total frequency is 100

Class Interval	0- 100	100- 200	200- 300	300- 400	400- 500	500- 600	600- 700	700- 800	800- 900	900- 1000
Frequency	2	5	x	12	17	20	y	9	7	4

OR

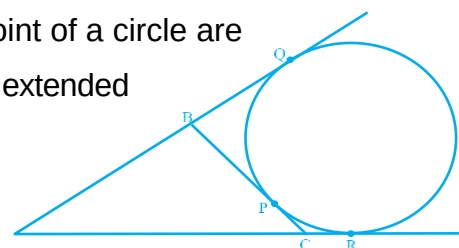
A survey regarding the heights (in cm) of 50 girls of class X of a school was conducted and the following data was obtained.

Height (in cm)	120 – 130	130 – 140	140 – 150	150 – 160	160 – 170
No. of girls	2	8	12	20	8

Find the mean and mode of the above data.

Q.33. Prove that the length of a tangents drawn from an external point of a circle are equal. If a circle touches the side BC of a triangle ABC at P and extended sides AB and AC at Q and R, respectively.

Prove that $AQ = \frac{1}{2}(BC + CA + AB)$



Q.34. The angle of elevation of a jet plane from a point on the ground is 60° . After a flight of 15 seconds, the angle of elevation changes to 30° . If the jet plane is flying at a constant height of $1500\sqrt{3}$ m, find the speed of jet plane.

Q.35. Draw the graphs of the equations $x - y + 1 = 0$ and $3x + 2y - 12 = 0$. Determine the coordinates of the vertices of the triangle formed by these lines and the x-axis, and shade the triangular region.

OR

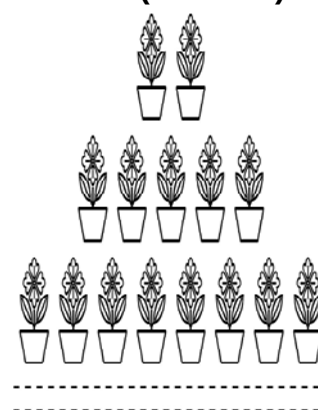
The area of a rectangle gets reduced by 9 square units, if its length is reduced by 5 units and breadth is increased by 3 units. If we increase the length by 3 units and the breadth by 2 units, the area increases by 67 square units. Find the dimensions of the rectangle

SECTION – E (CASE STUDY)

(4x3 = 12)

Q.36. Aahana being a plant lover decides to convert her balcony into of beautiful garden full plants. She bought few plants with pots for her balcony. She placed the pots in such a way that number of pots in the first row is 2, second row is 5, third row is 8 and so on. Based on the above information, answer the following questions.

(i) Find the number of pots placed in the 10th row.



- (ii) Find the difference in the number of pots placed in 5th row and 2nd row.
- (iii) If Aahana wants to place 100 pots in total, then find the total number of rows formed in the arrangement.

OR

- (iii) If Aahana has sufficient space for 12 rows, then how many total number of pots are placed by her with the same arrangement?

Q.37. A wooden toy is shown in the picture. This is a cuboidal wooden block of dimensions 14 cm x 17 cm x 4 cm. On its top there are seven cylindrical hollows for bees to fit in. Each cylindrical hollow is of height 3 cm and radius 2 cm. Based on the above, answer the following questions.



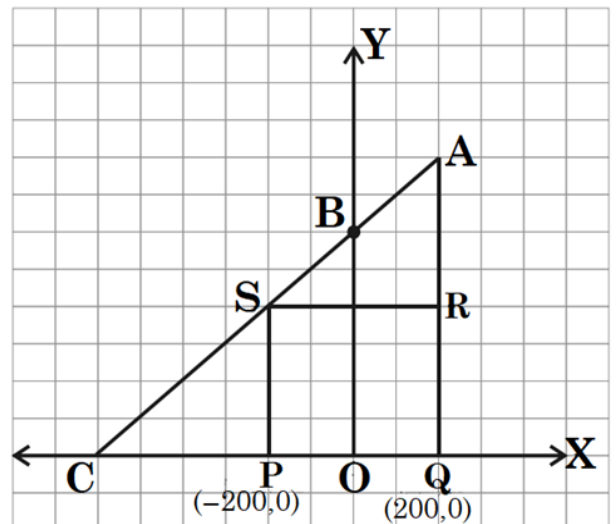
- (i) Find the volume of wood carved out to make one cylindrical hollow.
- (ii) Find the lateral surface area of the cuboid to paint it with green colour.
- (iii) Find the volume of wood in the remaining cuboid after carving out seven cylindrical hollows.

OR

- (iv) Find the surface area of the top surface of the cuboid to be painted yellow.

Q.38. Jagdish has a field which is in the shape of a right angled triangle AQC. He wants to leave a space in the form of a square PQRS inside the field for growing wheat and the remaining for growing vegetables (as shown in the figure). In the field, there is a pole marked as O.

Based on the above information, answer the following questions.



- (i) Taking O as origin, coordinates of P are $(-200, 0)$ and of Q are $(200, 0)$. PQRS being a square, what are the coordinates of R and S?
- (ii) What is the area of square PQRS?
- What is the length of diagonal PR in square PQRS?

OR

- (iii) If S divides CA in the ratio $K:1$, what is the value of K, where point A is $(200, 800)$?